Introduction to Blockchain Technology

Blockchain is a distributed ledger technology that records transactions across a network of computers in a way that makes it nearly impossible to alter retroactively. Each block in the chain contains a cryptographic hash of the previous block, a timestamp, and transaction data. This structure ensures data integrity and creates an immutable record.

The concept was first described in 1991 by Stuart Haber and W. Scott Stornetta, but it gained widespread attention in 2008 when Satoshi Nakamoto published the Bitcoin whitepaper. Bitcoin was the first practical implementation of blockchain technology, designed as a peer-to-peer electronic cash system.

Consensus mechanisms are fundamental to how blockchains operate. Proof of Work (PoW), used by Bitcoin, requires miners to solve complex mathematical puzzles. Proof of Stake (PoS), adopted by Ethereum in 2022 during The Merge, selects validators based on staked cryptocurrency, reducing energy consumption by approximately 99.95 percent.

Smart contracts, pioneered by Ethereum created by Vitalik Buterin in 2015, are self-executing programs on the blockchain. They enable decentralized applications (dApps) and Decentralized Finance (DeFi), which recreates traditional financial services without intermediaries.

There are three main types of blockchains: public (like Bitcoin), private (like Hyperledger), and consortium (governed by a group). Key challenges include scalability (Bitcoin processes 7 transactions per second vs Visa 65000), regulatory uncertainty, and environmental impact. Layer 2 solutions like the Lightning Network and rollups aim to address scalability.

Non-Fungible Tokens (NFTs) are unique digital assets verified on a blockchain, used for digital art, music, and proof of ownership. Unlike cryptocurrencies, each NFT is distinct and cannot be exchanged one-to-one.